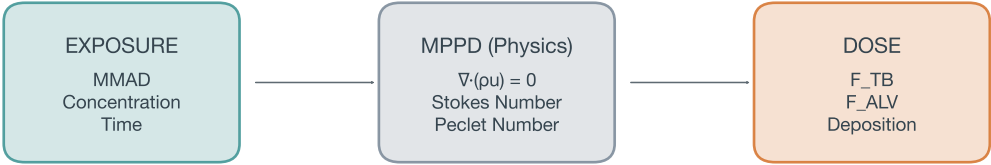
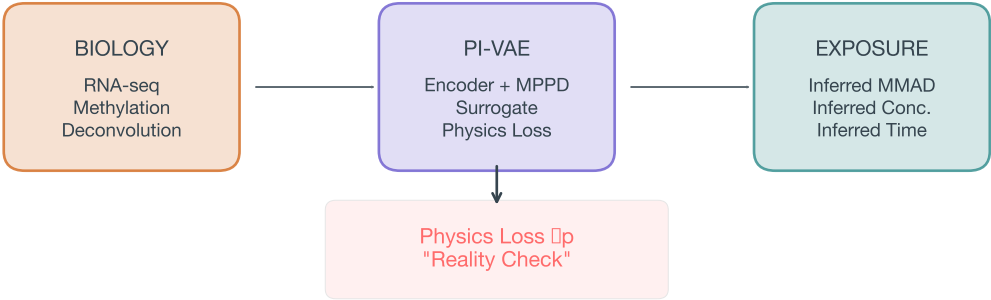


A THE FORWARD PROBLEM (Deterministic, Well-Posed)



B THE INVERSE PROBLEM (Ill-Posed □ Physics Constrained)



C

[Schematic: 3D Lung with Biomarker Gradients]
Trachea □ CYP1A1 | Alveoli □ MMP12